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Experiment 2: To Containerize Application using Docker

Aim:

To containerize application using Docker This experiment

maps to LO4.

LO4: Containerize, orchestrate and deploy applications using Docker, Docker Compose, Kubernetes and cloud infrastructure tools.

Theory:

Steps to install docker:

i. Navigate to the official Docker website (docker.com) and download the Docker Desktop installer for your operating system (Windows).

ii. Locate the downloaded installer file on your computer and double-click it to run the setup wizard.

iii. Follow the on-screen prompts. When asked, ensure the option to Use WSL 2 instead of Hyper-V is checked (for optimal performance on Windows).

iv. Wait for the installation to finish, click "Close," and restart your computer if the installer prompts you to do so.

v. After restarting, search for "Docker Desktop" in your system's Start menu and launch the application. Accept the service agreement to start the Docker engine.

vi. Open your terminal or PowerShell and type docker --version and press Enter. If it returns a version number, Docker is successfully installed and ready to use.

Docker:

Docker is an open-source platform that allows you to package an application and all its required dependencies into a single, standardized unit for software development. This ensures that the application runs exactly the same way regardless of where it is deployed—whether on your personal laptop, a testing environment, or a live production server. Instead of dealing with the old problem of "it works on my machine but not yours," Docker isolates the software so it behaves predictably everywhere.

Docker Images: A Docker image is a lightweight, standalone, and immutable (read-only) executable package. In theory, it is a static file that contains the complete set of instructions and components necessary to run a specific application.

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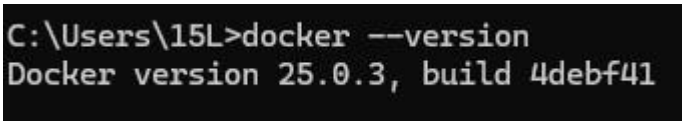
An image is built from multiple stacked layers, where each layer represents a specific instruction—such as installing a dependency, setting an environment variable, or copying application code. It encompasses the application code itself, the runtime environment, system tools, system libraries, and default configuration settings. Because it is read-only, an image cannot be changed once it is built.

Docker Containers: A container is the active, runtime execution of a Docker image. It is a standardized, isolated operating environment where the application actually runs and processes data.

In theory, when an image is executed, the Docker engine adds a thin, writable layer on top of the static image layers. This writable layer allows the active container to modify files, generate logs, and maintain a temporary state while it is running. Containers isolate the running software from the host operating system and from other containers, which ensures the application performs uniformly and predictably regardless of the underlying infrastructure.

Implementation:

1. docker --version



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2. docker search ubuntu

```
C:\Users\15L>docker search ubuntu
NAME                              DESCRIPTION                                     STARS     OFFICIAL
ubuntu                            Ubuntu is a Debian-based Linux operating sys…   17859     [OK]
ubuntu/squid                      Squid is a caching proxy for the Web. Long-t…   129
ubuntu/nginx                      Nginx, a high-performance reverse proxy & we…   141
ubuntu/bind9                      BIND 9 is a very flexible, full-featured DNS…   120
ubuntu/kafka                      Apache Kafka, a distributed event streaming …   62
ubuntu/zookeeper                  ZooKeeper maintains configuration informatio…   15
ubuntu/apache2                    Apache, a secure & extensible open-source HT…   104
ubuntu/prometheus                 Prometheus is a systems and service monitori…   80
ubuntu/jre                        Chiseled Java runtime based on Ubuntu. Long-…   25
ubuntu/dotnet-aspnet              Chiseled Ubuntu runtime image for ASP.NET a…   26
ubuntu/python                     A chiselled Ubuntu rock with the Python runt…   33
ubuntu/mysql                      MySQL open source fast, stable, multi-thread…   74
ubuntu/postgres                   PostgreSQL is an open source object-relation…   45
ubuntu/redis                      Redis, an open source key-value store. Long-…   26
ubuntu/dotnet-runtime             Chiselled Ubuntu runtime image for .NET apps…   24
ubuntu/dotnet-deps                Chiselled Ubuntu for self-contained .NET & A…   16
ubuntu/grafana                    Grafana, a feature rich metrics dashboard & …   12
ubuntu/memcached                  Memcached, in-memory keyvalue store for smal…   7
ubuntu/jdk                        Ubuntu rock for OpenJDK                         0
ubuntu/rabbitmq                   Ubuntu-based RabbitMQ image - an open source…   0
ubuntu/mimir                      Ubuntu ROCK for Mimir, a horizontally scalab…   0
ubuntu/loki                       Grafana Loki, a log aggregation system like …   2
ubuntu/traefik                    Ubuntu ROCK for Traefik, a modern HTTP rever…   3
ubuntu/opentelemetry-collector    This is a rock for the OpenTelemetry Collect…   0
ubuntu/alertmanager               Ubuntu Rock for Alertmanager. It handles ale…   1
```

3. search ubuntu --filter stars=100

```
C:\Users\15L>docker search --filter stars=100 ubuntu
NAME              DESCRIPTION                                     STARS     OFFICIAL
ubuntu            Ubuntu is a Debian-based Linux operating sys…   17859     [OK]
ubuntu/squid      Squid is a caching proxy for the Web. Long-t…   129
ubuntu/nginx      Nginx, a high-performance reverse proxy & we…   141
ubuntu/apache2    Apache, a secure & extensible open-source HT…   104
ubuntu/bind9      BIND 9 is a very flexible, full-featured DNS…   120
```

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4. docker search postgres

```
C:\Users\15L>docker search postgres
NAME                      DESCRIPTION                                     STARS     OFFICIAL
postgres                  The PostgreSQL object-relational database sy…   14969     [OK]
cimg/postgres                                                             9
circleci/postgres         The PostgreSQL object-relational database sy…   35
kasmweb/postgres          Postgres image maintained by Kasm Technologi…   6
elestio/postgres          Postgres, verified and packaged by Elestio      2
ubuntu/postgres           PostgreSQL is an open source object-relation…   45
cleanstart/postgres       Secure by Design, Built for Speed, Hardened …   0
artifacthub/postgres                                                      0
geokrety/postgres         Postgres with postgis + quantile and amqp ex…   0
corpusops/postgres        https://github.com/corpusops/docker-images/     0
rootpublic/postgres                                                       0
dockette/postgres         My PostgreSQL image with tunning and preinst…   1
vulhub/postgres                                                           2
wayofdev/postgres                                                         0
pgrouting/postgres         Postgres Docker images with PostGIS and dep…   0
uselagoon/postgres                                                        0
openeuler/postgres                                                        0
clarinpl/postgres                                                         0
mosipdev/postgres                                                         0
supabase/postgres         Unmodified Postgres with some useful plugins…   78
brimstone/postgres        postgres image with traefik-cert support        0
trainlineeurope/postgres  Extended version of official Postgres https:…   0
blacklabelops/postgres    Postgres Image for Atlassian Applications       4
fredboat/postgres         PostgreSQL 10.0 used in FredBoat's docker-co…   1
sushantdhiman/postgres    Postgres Image for Sequelize                    2
```

5. docker ps

```
C:\Users\15L>docker images
REPOSITORY              TAG       IMAGE ID       CREATED        SIZ
ubuntu                  latest    de7345b16e94   8 days ago     100
ubuntu                  24.04     ef91e4b15da8   4 weeks ago    78.
sample-flask-app        v1        ad2b8021fa04   4 months ago   154
redis                   latest    c234a24851ba   4 months ago   140
app                     latest    f881ae5ad2c9   4 months ago   161
myapp                   latest    a5a22179b024   4 months ago   161
new_image_name          redis3    c133193b1b3e   4 months ago   138
redis                   <none>    f0acdeb5ced3   4 months ago   140
instrumentisto/nmap     latest    977143425e3d   5 months ago   42.
node                    latest    8bbb9d1bc633   2 years ago    1.1
t13-49                  latest    acb00e2365d5   2 years ago    912
hello-world             latest    7e4af798f449   2 years ago    13.
<none>                  <none>    9c469d82cc74   2 years ago    13.
ubuntu                  <none>    ca2b0f26964c   2 years ago    77.
redis                   <none>    170a1e90f843   2 years ago    138
redis                   <none>    d1397258b209   2 years ago    138
hello-world             <none>    d2c94e258dcb   3 years ago    13.
vulnerables/web-dvwa    latest    ab0d83586b6e   7 years ago    712
```

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6. docker images

7. docker run -it ubuntu

```
C:\Users\15L>docker run -it ubuntu /bin/bash
root@a1a5840d97aa:/# ls
bin  boot  dev  etc  home  lib  lib64  media  mnt  opt  proc  root  run  sbin  srv  sys  tmp  usr  var
root@a1a5840d97aa:/# exit
exit
```

8. docker build -t flask-app .

```
C:\Users\15L\docker-demo>docker build -t flask-app .
[+] Building 8.5s (10/10) FINISHED
 => [internal] load build definition from Dockerfile
 => => transferring dockerfile: 212B
 => [internal] load metadata for docker.io/library/python:3.12-slim
 => [internal] load .dockerignore
 => => transferring context: 2B
 => [1/5] FROM docker.io/library/python:3.12-slim@sha256:57cd7c3a7a273101a6485ba99423ee568157882804b1124b4dd04266317710de
 => [internal] load build context
 => => transferring context: 297B
 => CACHED [2/5] WORKDIR /app
 => [3/5] COPY requirements.txt .
 => [4/5] RUN pip install --no-cache-dir -r requirements.txt
 => [5/5] COPY . .
 => exporting to image
 => => exporting layers
 => => writing image sha256:6a7ecbf32813e8784464a013438b8fdfa597f1de748a3e3e774d02c127aa4c6b
 => => naming to docker.io/library/flask-app

What's Next?
  1. Sign in to your Docker account → docker login
  2. View a summary of image vulnerabilities and recommendations → docker scout quickview

C:\Users\15L\docker-demo>docker run -d -p 5000:5000 flask-app
cfd496e8428243a75529cbdbf8725a8ead3668ccc4c57d72aeabef4f09470d60

C:\Users\15L\docker-demo>docker ps
CONTAINER ID   IMAGE       COMMAND          CREATED          STATUS       PORTS                    NAMES
cfd496e84282   flask-app   "python app.py"   11 seconds ago   Up 6 hours   0.0.0.0:5000->5000/tcp   practical_cohen

C:\Users\15L\docker-demo>docker logs <container_id>
The syntax of the command is incorrect.

C:\Users\15L\docker-demo>docker logs <container_id>
The syntax of the command is incorrect.

C:\Users\15L\docker-demo>docker logs <cfd496e84282>
The syntax of the command is incorrect.

C:\Users\15L\docker-demo>docker logs cfd496e84282
 * Serving Flask app 'app'
 * Debug mode: off
WARNING: This is a development server. Do not use it in a production deployment. Use a production WSGI server instead.
 * Running on all addresses (0.0.0.0)
 * Running on http://127.0.0.1:5000
 * Running on http://172.17.0.2:5000
Press CTRL+C to quit
```

9. docker run -d -p 5000:5000 flask-app

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```
C:\Users\15L\docker-demo>docker run -d -p 5000:5000 flask-app
cfd496e8428243a75529cbdbf8725a8ead3668ccc4c57d72aeabef4f09470d60
```

10. docker ps

```
C:\Users\15L\docker-demo>docker rmi flask-app
Untagged: flask-app:latest
Deleted: sha256:6a7ecbf32813e8784464a013438b8fdfa597f1de748a3e3e774d02c127a
```

11. docker logs cfd496e84282

12. docker stop cfd496e84282

```
C:\Users\15L\docker-demo>docker stop cfd496e84282
cfd496e84282
```

13. docker rm cfd496e84282

```
C:\Users\15L\docker-demo>docker rm cfd496e84282
cfd496e84282
```

14. docker rmi flask-app

```
C:\Users\15L\docker-demo>docker ps
CONTAINER ID   IMAGE       COMMAND           CREATED          STATUS       PORTS                    NAMES
cfd496e84282   flask-app   "python app.py"   11 seconds ago   Up 6 hours   0.0.0.0:5000->5000/tcp   practic
```

```
C:\Users\15L\docker-demo>docker logs cfd496e84282
 * Serving Flask app 'app'
 * Debug mode: off
WARNING: This is a development server. Do not use it in a production deployment. Use a production WSGI server
 * Running on all addresses (0.0.0.0)
 * Running on http://127.0.0.1:5000
 * Running on http://172.17.0.2:5000
Press CTRL+C to quit
```

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Conclusion:

Conclusion

In this experiment, the fundamental concepts of Docker and application containerization were successfully understood and implemented by containerizing a Flask web application. The theoretical study established how Docker enables applications to run in isolated, portable, and consistent environments, while highlighting the importance of Dockerfiles, Docker images, and Docker containers in the application deployment process.

During the practical implementation, a complete workflow for building and running a containerized Flask application was successfully performed. The experiment demonstrated how to:

- Create a Dockerfile to define the application environment.
- Specify project dependencies using a requirements.txt file.
- Build a Docker image from the application source code.
- Run the Docker container with appropriate port mapping to make the application accessible through a web browser.
- Verify the application's execution and manage the container using Docker commands. Overall, this experiment provided practical knowledge of packaging a Python Flask application into a Docker container, enabling consistent deployment across different systems without dependency or environment-related issues. The skills acquired through this experiment form a strong foundation for developing, deploying, and managing containerized applications in modern software development and cloud environments.